EXPERIMENT NUMBER: 10

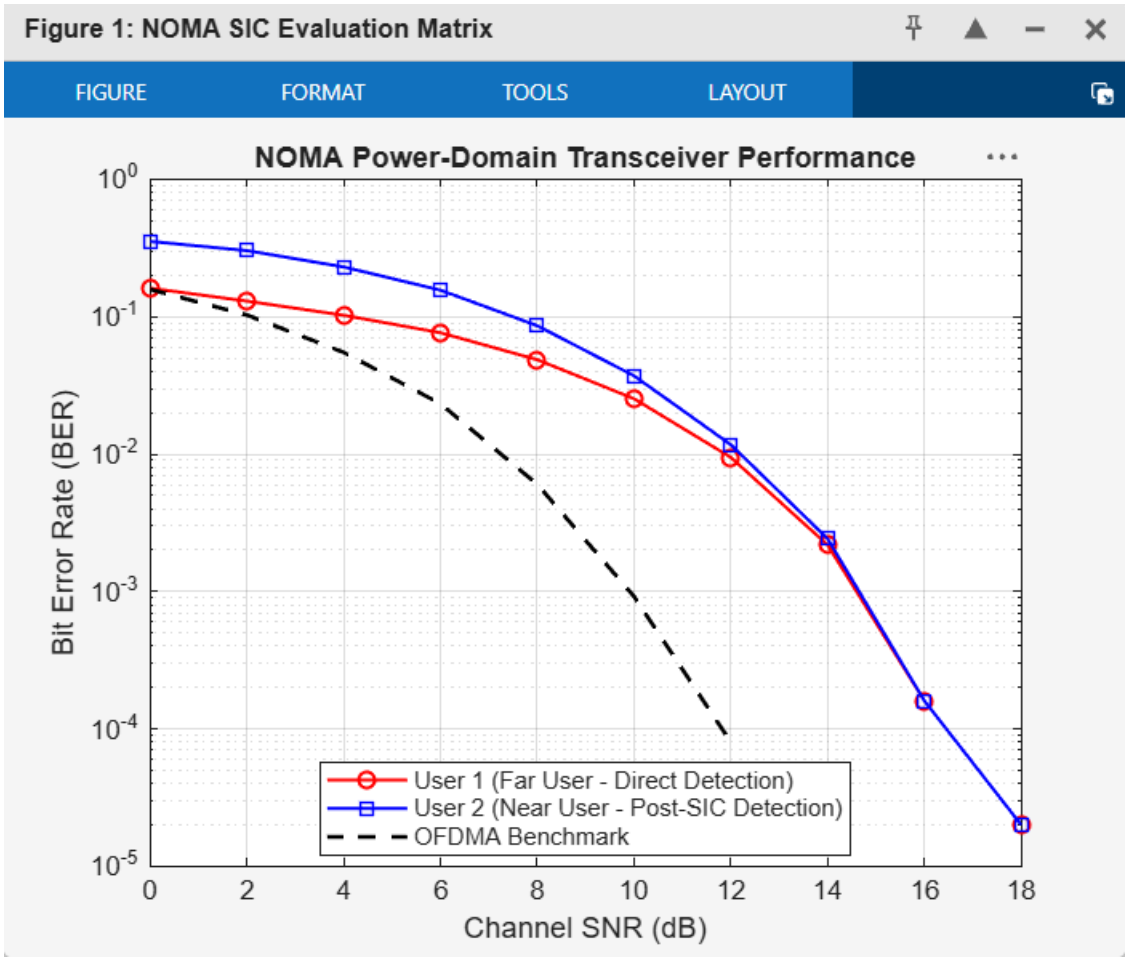
Simulation and Performance Evaluation of NOMA and Successive Interference Cancellation

Matlab Program

```
clc; clear; close all;
%% 1. Input Parameters Setup
N_bits = 50000;                % Data length per SNR point
SNR_dB = 0:2:20;               % SNR operating range
alpha1 = 0.75;                 % Far user (User 1) power allocation
alpha2 = 1 - alpha1;           % Near user (User 2) power allocation
ber_User1 = zeros(length(SNR_dB), 1);
ber_User2 = zeros(length(SNR_dB), 1);
ber_OFDMA = zeros(length(SNR_dB), 1);
%% 2. Data Processing Execution Loop
for s = 1:length(SNR_dB)
    snr_linear = 10^(SNR_dB(s)/10);
    noise_variance = 1 / snr_linear;
% Source Bits Generation
    bits_u1 = randi([0 1], N_bits, 1);
    bits_u2 = randi([0 1], N_bits, 1);
% BPSK Constellation Mapping
    x1 = 2*bits_u1 - 1;
    x2 = 2*bits_u2 - 1;
% Power-Domain NOMA Superposition
    x_noma = sqrt(alpha1)*x1 + sqrt(alpha2)*x2;
% AWGN Channel
    noise = sqrt(noise_variance/2) * ...
        (randn(N_bits, 1) + 1i*randn(N_bits, 1));
        y_noma = real(x_noma + noise);
%% 3. User 1 Direct Detection
```

```
    dec_u1 = y_noma > 0;
est_x1 = 2*double(dec_u1) - 1;
ber_User1(s) = sum(bits_u1 ~= dec_u1) / N_bits;
%% 4. Successive Interference Cancellation (SIC)
    y_sic = y_noma - sqrt(alpha1)*est_x1;
    dec_u2 = y_sic > 0;
    ber_User2(s) = sum(bits_u2 ~= dec_u2) / N_bits;
%% 5. Baseline OFDMA Evaluation
    noise_ofdma = sqrt(noise_variance) * randn(N_bits, 1);
    y_ofdma = x1 + noise_ofdma;
    dec_ofdma = y_ofdma > 0;
    ber_OFDMA(s) = sum(bits_u1 ~= dec_ofdma) / N_bits;
end
%% 6. Performance Graph
figure('Name', 'NOMA SIC Evaluation Matrix');
semilogy(SNR_dB, ber_User1, 'ro-', 'LineWidth', 1.2, 'MarkerSize', 6);
hold on;
semilogy(SNR_dB, ber_User2, 'bs-', 'LineWidth', 1.2, ‘MarkerSize', 6);
semilogy(SNR_dB, ber_OFDMA, 'k--', 'LineWidth', 1.5);
grid on;
xlabel('Channel SNR (dB)');
ylabel('Bit Error Rate (BER)');
title('NOMA Power-Domain Transceiver Performance');
legend('User 1 (Far User - Direct Detection)', ...
       'User 2 (Near User - Post-SIC Detection)', ...
       'OFDMA Benchmark', 'Location', 'best');
hold off;
```

Result


Figure 1: NOMA SIC Evaluation Matrix
FIGURE
FORMAT
TOOLS
LAYOUT
NOMA Power-Domain Transceiver Performance
Bit Error Rate (BER)
Channel SNR (dB)
User 1 (Far User - Direct Detection)
User 2 (Near User - Post-SIC Detection)
OFDMA Benchmark